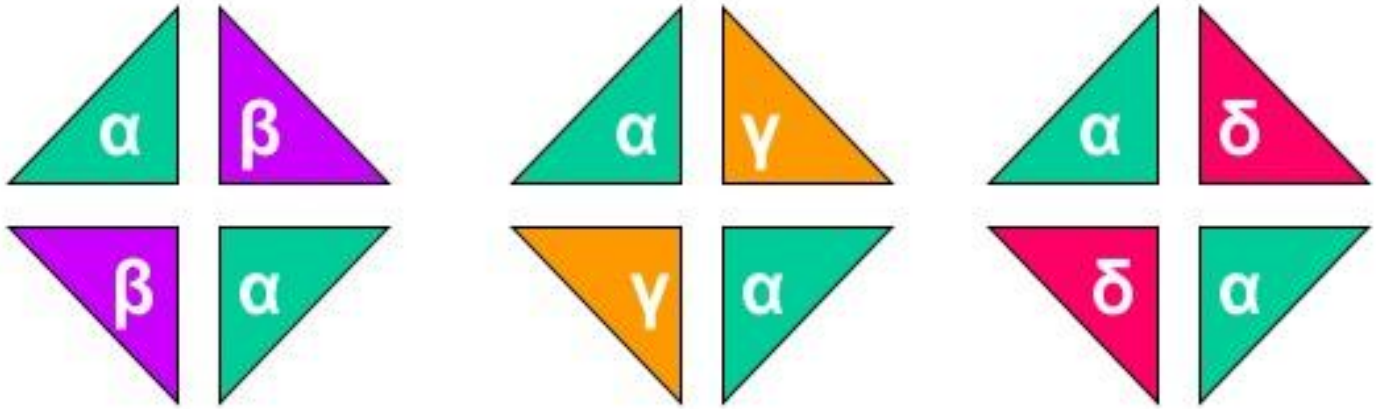
HEMOGLOBINOPATHIES

Hemoglobins in normal adults



HbA HbF HbA_2

HEMOGLOBINOPATHIES

Definition:

" An inherited mutation of the globin genes leading to a qualitative or quantitative abnormality of globin synthesis".

OR

"Hemoglobinopathies are a group of genetic disorders caused by **production of a structurally abnormal hemoglobin molecule**; **synthesis of insufficient quantities of normal hemoglobin**; or, **rarely, both"**.

HEMOGLOBINOPATHIES

Qualitative hemoglobinopathy: production of hemoglobin with an altered amino acid sequence

- Sickle cell anemia (HbS)
- hemoglobin C disease (HbC)
- hemoglobin SC disease (HbS + HbC = HbSC)

Quantitative hemoglobinopathy: decreased production of normal hemoglobin

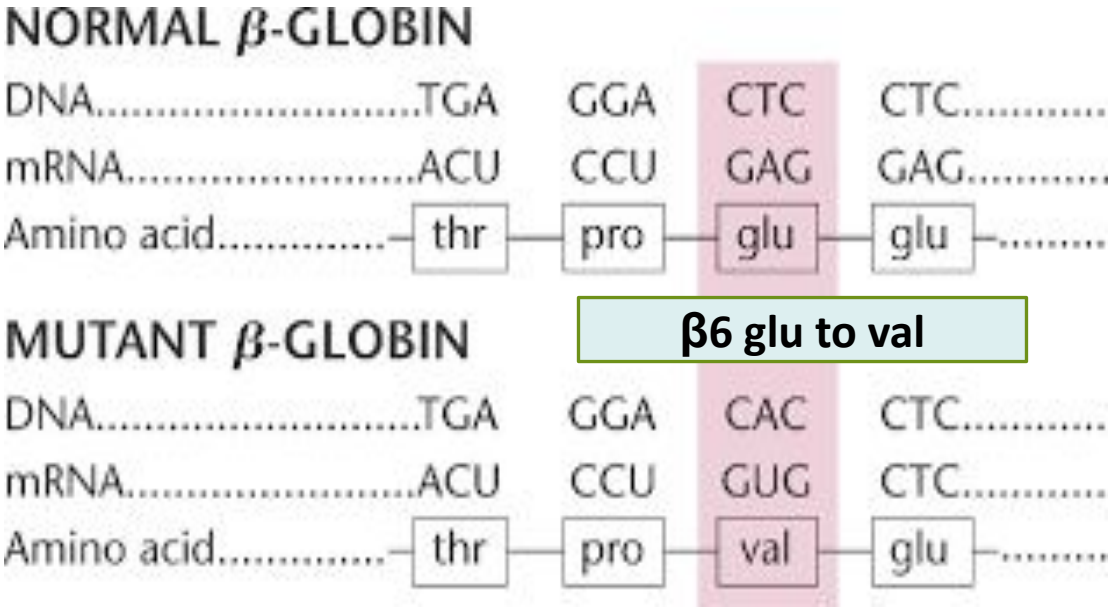
- Thalassemias

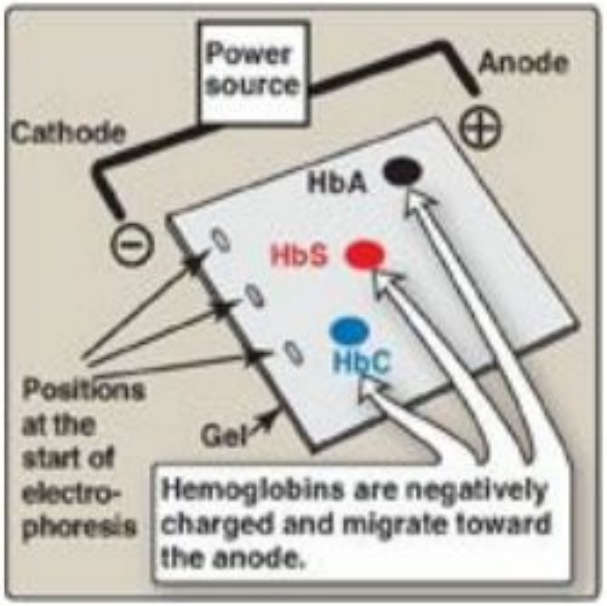
- **Abnormalities of hemoglobin structure cause hemolytic anemias that can have severe clinical consequences.**

HEMOGLOBIN S DISORDER (Sickle cell anemia)

General considerations

- An **autosomal recessive** genetic disorder, caused by a single nucleotide substitution in the gene for β-globin that code for synthesis of the β chains of the globin molecules.
- The mutant β-globin chain is designated βs, and the resulting hemoglobin, α2βS 2, is referred to as HbS.
- It occurs primarily in the African American population.





Sickle cell anemia

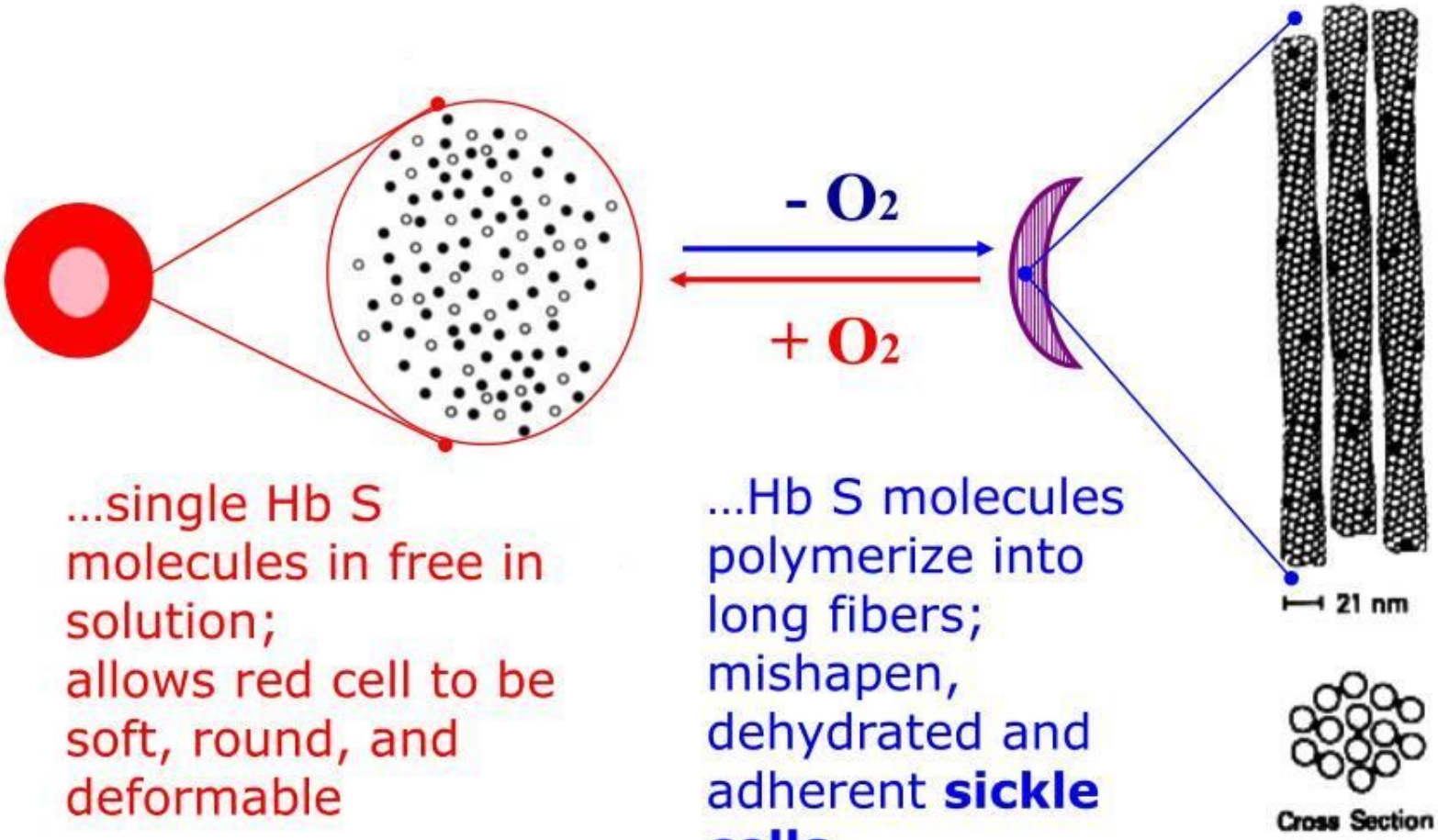
- Sickle cell anemia, the **most common** of the RBC sickling diseases.
- An infant does not begin showing symptoms of the disease until sufficient HbF has been replaced by HbS.
- Hemoglobin S polymerizes at low oxygen tension, forming **'Tactoids '** (a network of insoluble fibrous polymers) that distort the shape of red cells.
- Sickle shape red cells, repeated sickling episodes stiffen red cell membranes, making affected cells more subject to hemolysis.
- Small blood vessel occlusion and tissue infarction that results in "**Vaso-occlusive painful crises" (severe pain in limbs, back, chest and abdomen) often precipitated by infection or dehydration.**

Pathophysiology of SCD

In a red blood cell containing mostly Hb S…

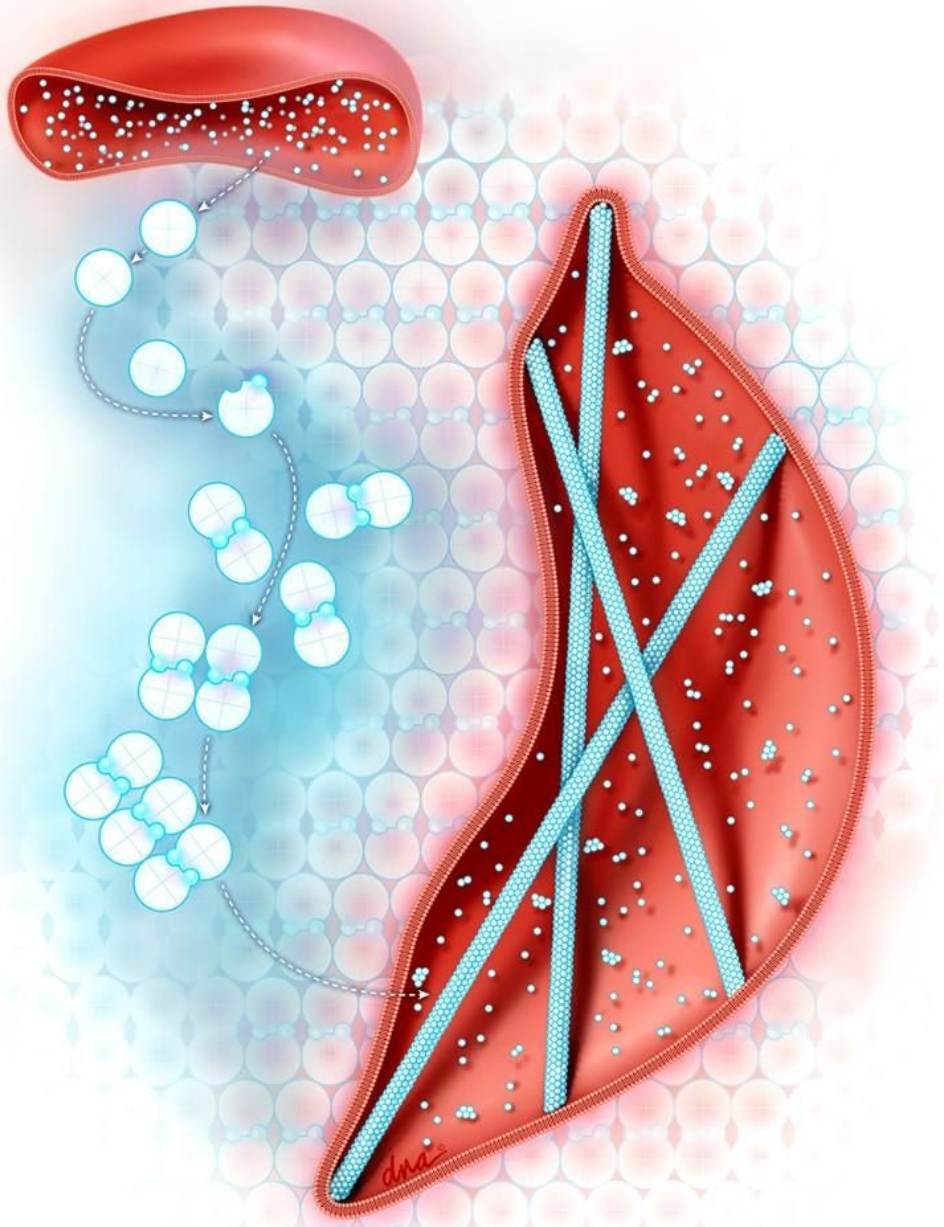
When oxygenated…

When deoxygenated



…single Hb S molecules in free in solution;
allows red cell to be soft, round, and deformable

…Hb S molecules polymerize into long fibers; mishapen, dehydrated and adherent **sickle cells.**



- Repeated infarctions in the lungs and spleen; the spleen is characteristically congested and enlarged in childhood, but becomes progressively smaller through repeated infarcts and fibrosis (autosplenectomy).

- Severe chronic hemolytic anemia with associated hyperbilirubinemia.

- Increased susceptibility to infections, usually beginning in infancy.

- Other symptoms include acute chest syndrome, stroke, splenic and renal dysfunction, and bone changes due to marrow hyperplasia.

Normal and Sickled Red Blood Cells in Blood Vessels

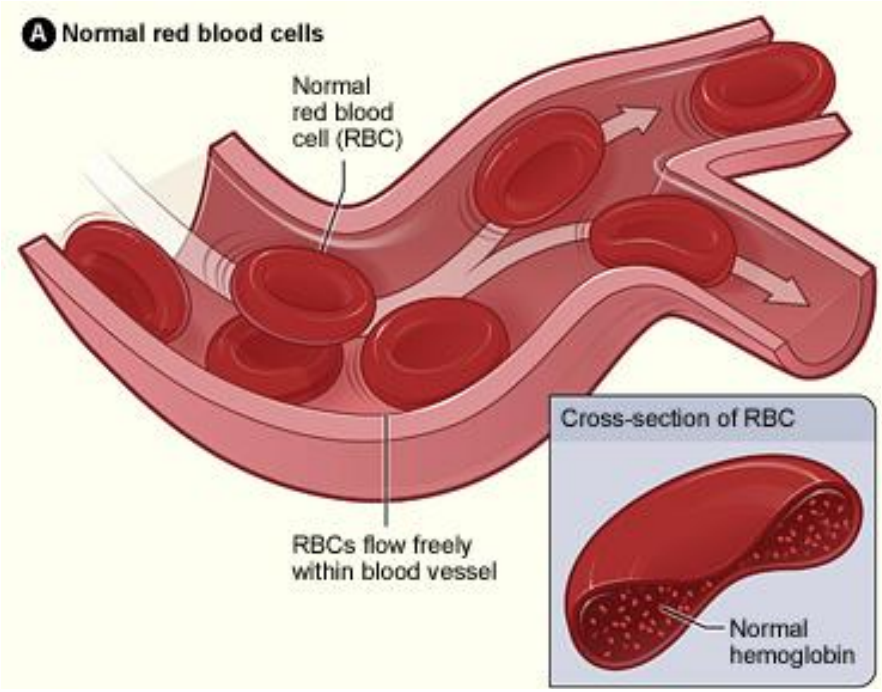
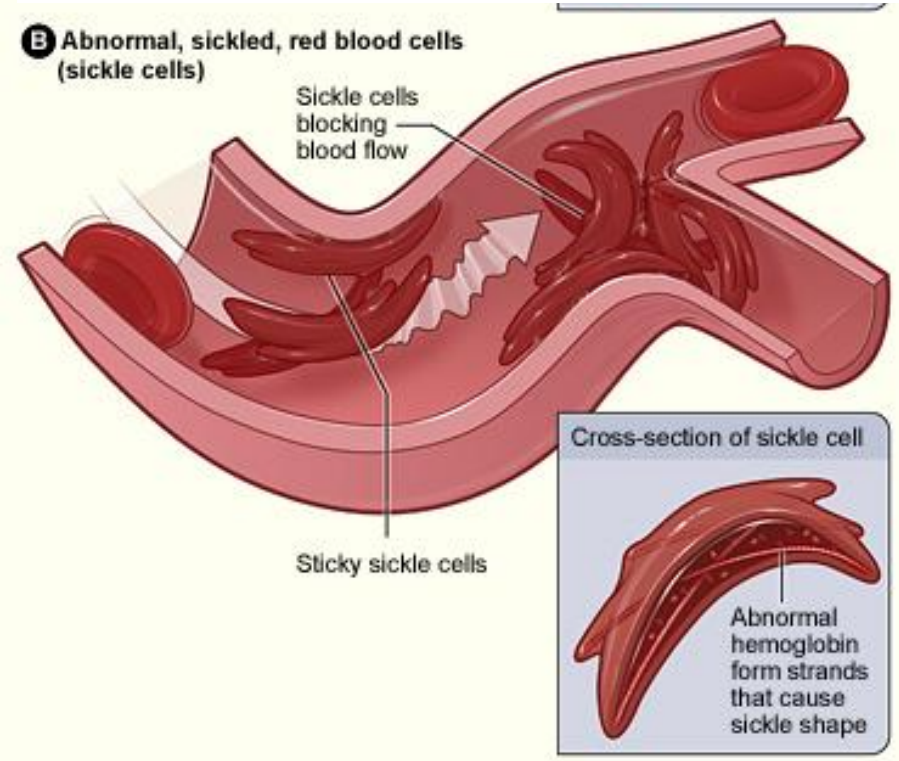


Figure A shows normal red blood cells flowing freely in a blood vessel. The inset image shows a cross-section of a normal red blood cell with normal hemoglobin.

Figure B shows abnormal, sickled red blood cells clumping and blocking the blood flow in a blood vessel. The inset image shows a cross-section of a sickled red blood cell with abnormal strands of hemoglobin.



Source from http://www.nhlbi.nih.gov/health/dci/Diseases/Sca/SCA_WhatIs.html

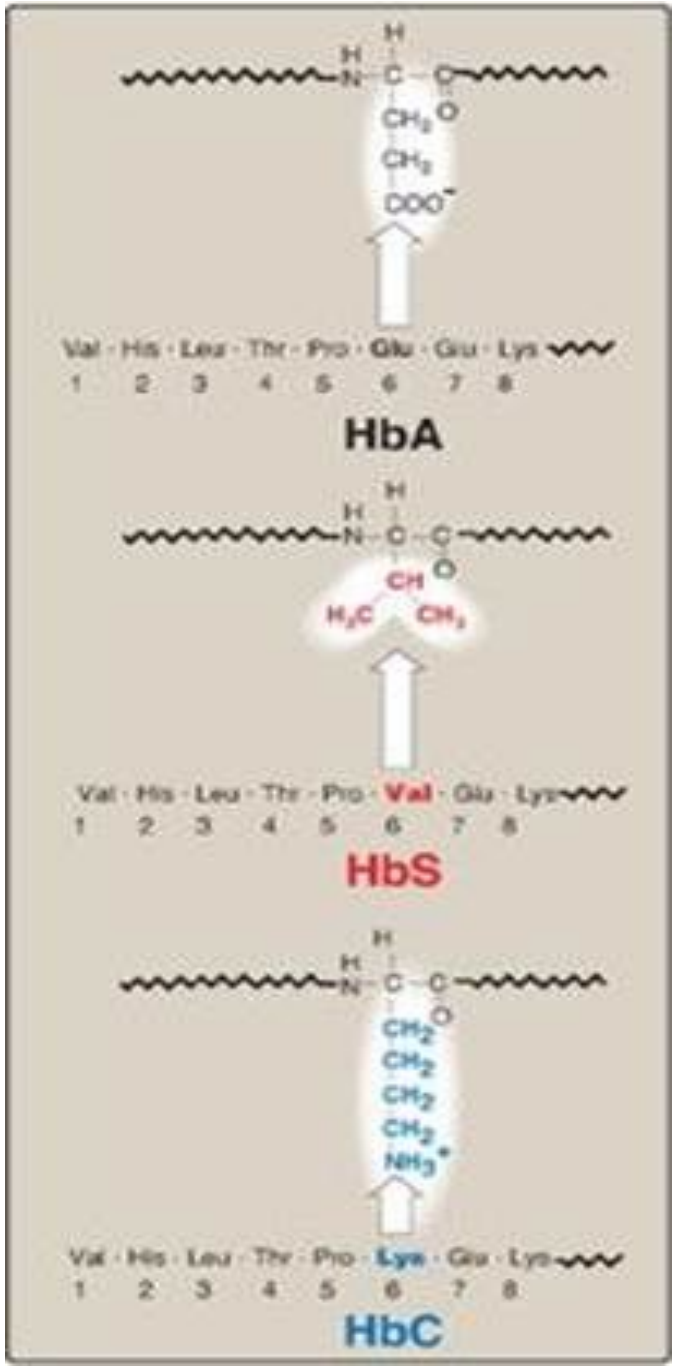
Sickle cell trait

- **Hemoglobin S in the heterozygous form (**have one normal and one sickle cell gene, the blood cells of such heterozygotes contain both HbS and HbA) **leads to sickle cell trait, which is generally without clinical consequence.**

- Heterozygotes for the sickle cell gene are less susceptible to the severe malaria caused by the parasite Plasmodium falciparum.

Hemoglobin C disease

- Like HbS, HbC is a hemoglobin variant that has a single amino acid substitution in the sixth position of the β-globin chain.
- A lysine is substituted for the glutamate
- Rare patients homozygous for HbC generally have a relatively mild, chronic hemolytic anemia.
- These patients do not suffer from infarctive crises, and no specific therapy is required.



Thalassemia

Definition:
a blood disorder passed down through families (inherited) in which the body makes an abnormal form of hemoglobin, the protein in red blood cells that carries oxygen

Formal Names:
Mediterranean anemia; Cooley's anemia; Beta thalassemia; Alpha thalassemia

This disease is inhereited

Starts from birth, you can get tested for it.

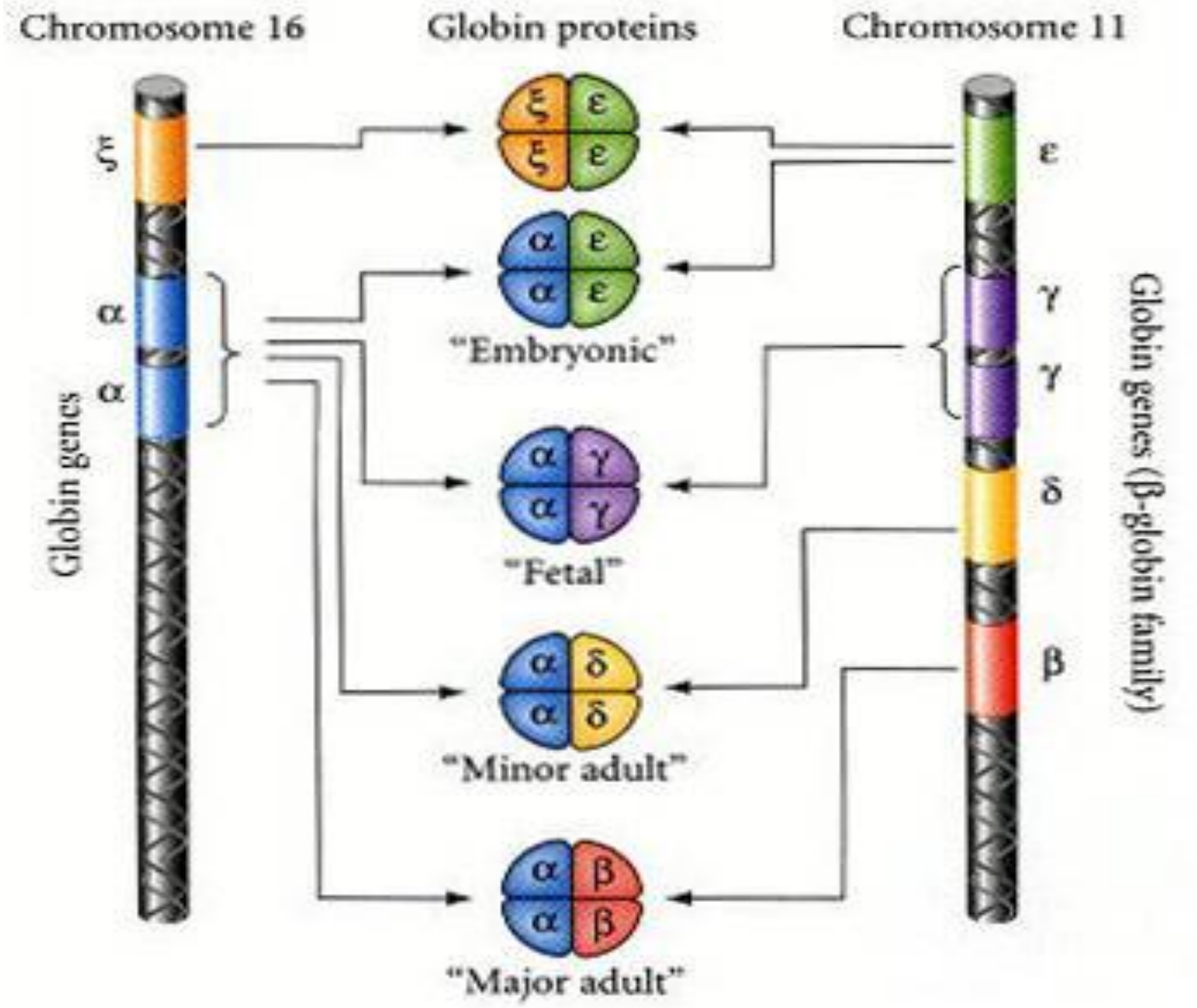
Affects the Bone Marrow

Treatment involves blood transfusions that must be given every 4 to 6 weeks to sustain life.

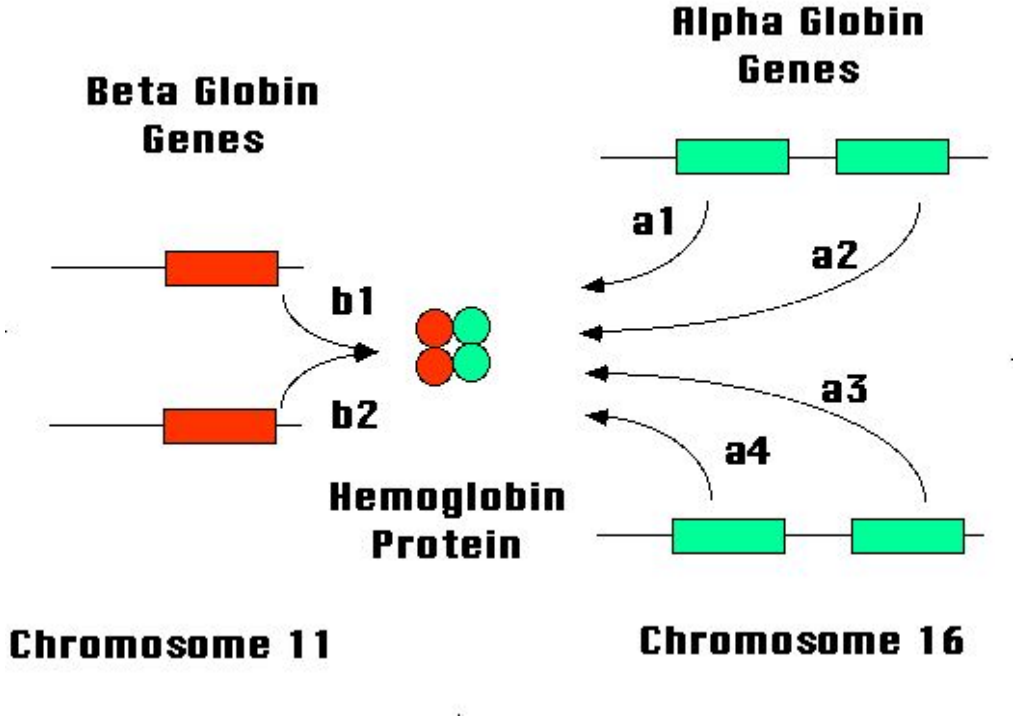
Can be acute or chronic, depends whether it's major or minor

THALASSEMIA (thal; α and β)

- This heterogeneous group of genetic disorders is characterized by deficient production of either α- or β-globin chains of hemoglobin.
- **Heme synthesis is unaffected.**
- Epidemiology:
  - Autosomal recessive disorder.
  - They are widespread throughout the world, occurring with high frequency in:
    - α- Thal is common in southeast Asia and in African Americans.
    - B- Thal is common in Mediterranean areas and in the United states.


Chromosome 16
Globin proteins
Chromosome 11
ζ
α
α
Globin genes
ε
γ
γ
δ
β
Globin genes (β-globin family)
"Embryonic"
"Fetal"
"Minor adult"
"Major adult"

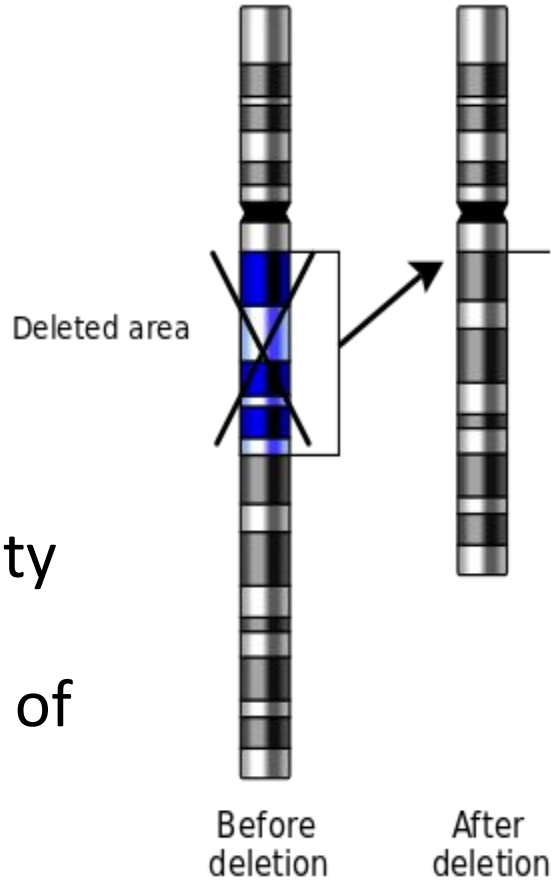
- The two chromosomes #11 have one beta globin gene each (for a total of two genes). The two chromosomes #16 have two alpha globin genes each (for a total of four genes).

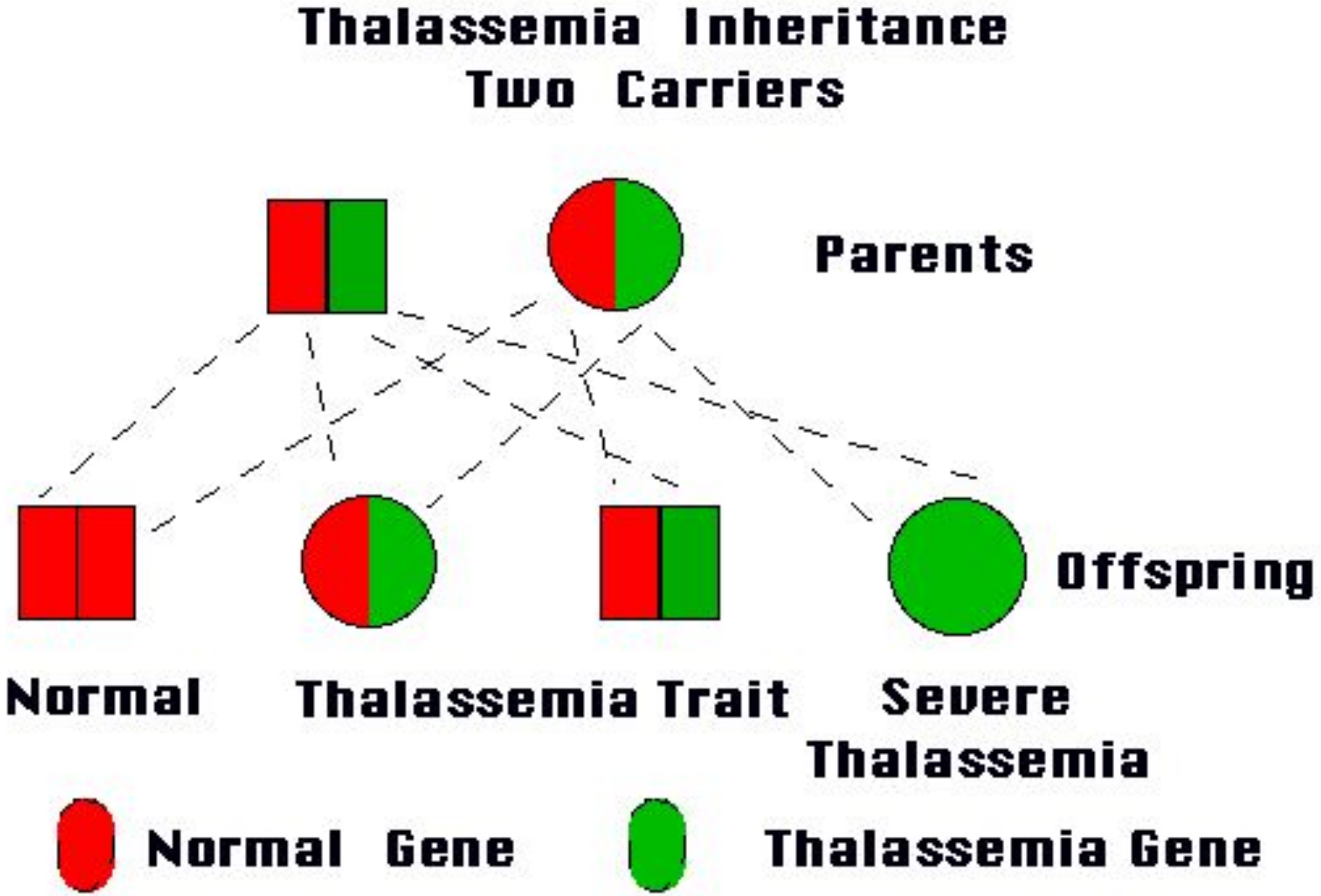


TYPES

- The **α-** and **β-talassemias** further classified into those producing no product (α_0, β_0) and those producing reduced amounts of product (α+, β+).

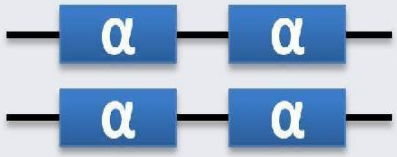
- A thalassemia can be caused by a variety of mutations, including entire gene deletions, or substitutions or deletions of one to many nucleotides in the DNA.



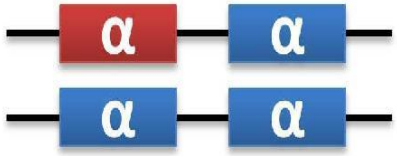

Thalassemia Inheritance
Two Carriers
Parents
Offspring
Normal
Thalassemia Trait
Severe
Thalassemia
Normal Gene
Thalassemia Gene

Alpha-thalassemia Genetics and Clinical Consequences

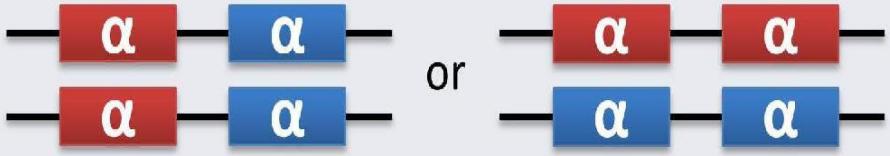
Normal



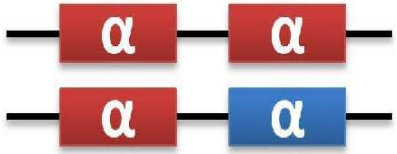
Carrier: Asymptomatic
No abnormalities



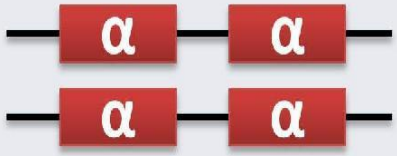
α-thal minor: Asymptomatic
Mild microcytic anemia



Hb H Disease: Symptomatic
Hemolytic and Microcytic anemia
Splenomegaly



Incompatible with Life
Hydrops Fetalis





Type of Hemoglobin

- Hb A $\alpha_2\beta_2$
- Hb A2 $\alpha_2\delta_2$
- Hb F $\alpha_2\gamma_2$
- Hb H β_4
- Hb Bart's γ_4

β-Thalassemias:

- **Decrease in β-globin chain synthesis**
  - **β^+ : reduced beta-globin chain synthesis,** β-thalassemia trait (β-thalassemia minor)
  - **β^0 : no beta-globin chain synthesis,** β-thalassemia major (Cooley's anemia)

- **More than 150 point mutations and several deletional mutations have been identified within and around the beta-globin chain gene.**

β – Thal Minor (beta^{+}-thalassaemia)

- Heterozygous inheritance
- Mild microcytic anemia
- Mild protective effect against falciparum malaria
- Decreased MCV, Hb, and Hct
- Normal serum ferritin
- On Hb electrophoresis
  - Decreased HbA
  - Increased HbA2(key diagnostic clue) and HbF

Beta-thalassemia minor with microcytes and target cells



β – Thal Major (beta^0-thalassaemia)

- **Severe hemolytic anemia**
  - **RBCs with α-chains inclusions are removed by macrophages in the spleen (marked spleenomegaly)**
  - **Ineffective erythropoiesis**
  - **Extramedullary hematopoiesis and distortion of skull, fascial bones and long bones because of erythroid marrow expansion**
  - **RBC morphology: marked microcytosis, hypochromic, target cells and extensive changes in size and shape**
  - **Increased HbF and HbA2**
  - **Generalized hemosiderosis.**

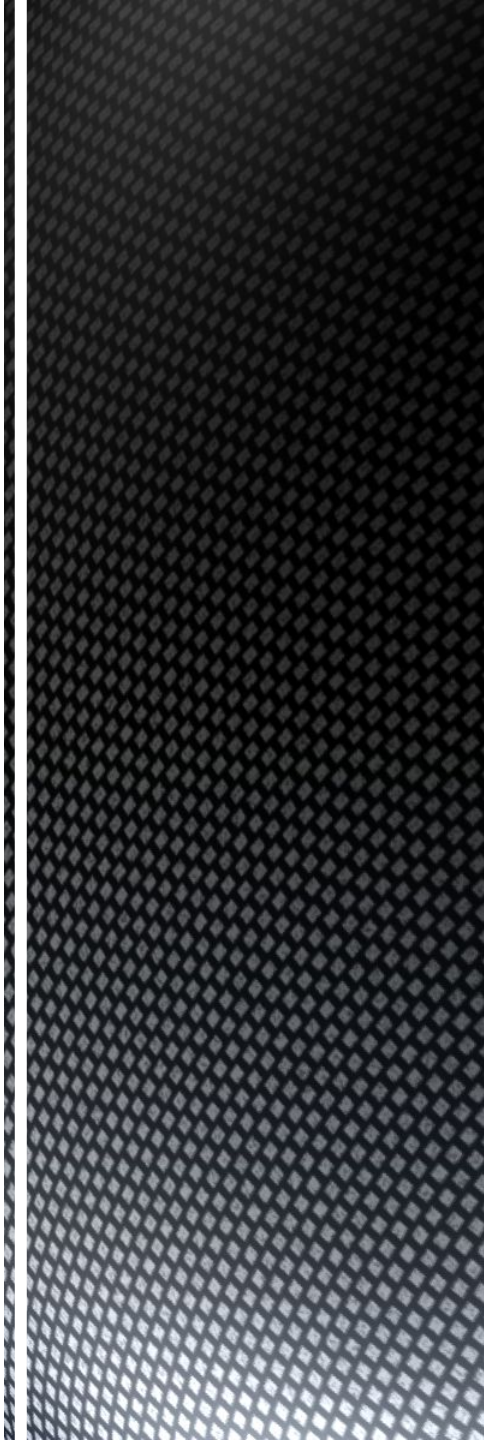

© CD-ROM ILLUSTRATED LECTURE NOTES ON TROPICAL MEDICINE



Methemoglobinemia

- Oxidation of the heme iron in hemoglobin to the ferric (Fe3+) state forms methemoglobin, which cannot bind O2.
- Causes:
  - Certain drugs, such as nitrates
  - Endogenous products such as reactive oxygen species
  - Inherited defects
  - Deficiency of methemoglobin reductase, the enzyme responsible for the conversion of methemoglobin (Fe3+) to hemoglobin (Fe2+)
- The methemoglobinemia are characterized by “chocolate cyanosis” (a brownish blue coloration of the skin and mucous membranes and brown-colored blood) as a result of the dark-colored HbM.
- Symptoms are related to the degree of tissue hypoxia and include anxiety, headache, and dyspnea. In rare cases, coma and death can occur.

Case Study



Clinical presentation:

A 2-year-old child is brought to the hospital with:

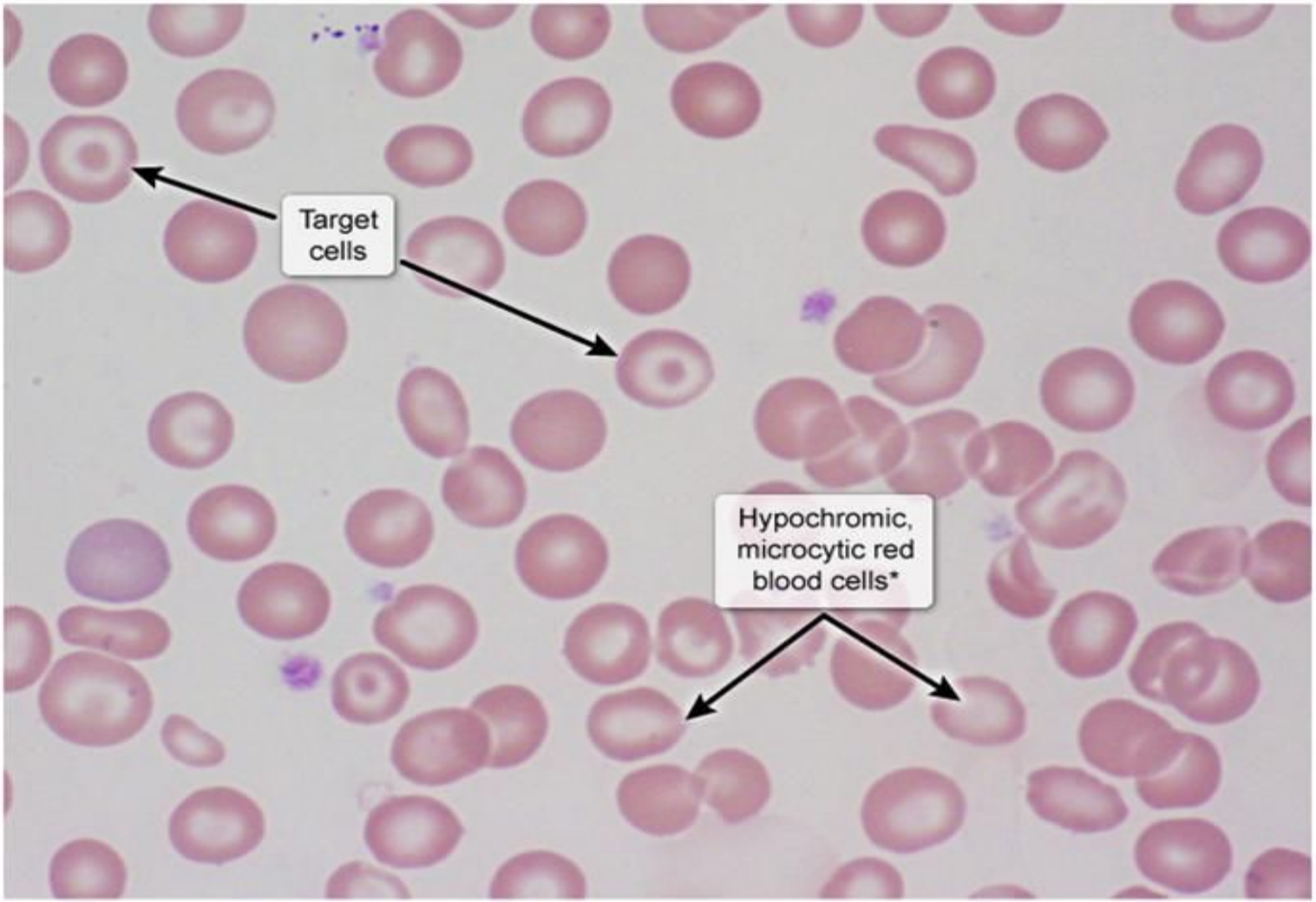
-Severe pallor and weakness.

- Failure to thrive.
- Abdominal distension.
- Enlarged spleen and liver (hepatosplenomegaly).
- History of repeated blood transfusions.

Laboratory findings:

- Hemoglobin: 5 g/dL.
- MCV: Markedly decreased.
- MCH: Decreased.
- Peripheral smear: Microcytic, hypochromic RBCs, target cells and nucleated RBCs.
- Hb electrophoresis: Increased HbF, increased HbA2, absent or very low HbA.

Beta-thalassemia



*Pale, small red blood cells.

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DIAGNOSIS

Beta-thalassemia major (Cooley's anemia)


Thank you!